REFERENCE DRAWINGS

GENERAL NOTES:

A. GENERAL NOTES, GENERAL HOLDS, SYMBOLS AND DETAILS REFER TO P&IDs 220900BPFM03500 TO 03501
B. ALL INSTRUMENT TAGS ON THIS P&ID SHALL BE PREFIXED BY 011900 UNLESS OTHERWISE INDICATED.
C. FOR INTERLOCK DETAILS REFERT TO C&E DOC. NO220900BPEQ03470.
D. FOR HEMP RECCOMENDATIONS REFER TO DOCUMENT 220901DPRI85111.

NOTES:

1. INSTRUSIVE TYPE PIG SIGNALLER TO BE INSTALLED ON MINOR BARREL AT MIN DISTANCE FROM THE REDUCER.
2. LAUNCHING/RECEIVING TRAP TO BE PROVIDED WITH MECHANICAL INTERLOCK SYSTEM DOOR THAT WILL PREVENT DOOR OPENING UNTIL THE PIG TRAP IS ISOLATED AND FULLY DEPRESSURIZED.
3. FOR BI-DIRECTIONAL PIG TRAPS, A SINGLE KICKER LINE COULD BE LOCATED APPROXIMATELY HALF WAY ALONG THE MAJOR BARREL.
4. INTELLIGENT PIG TO BE PROVIDED.
5. VALVE SHALL ALLOW INTEGRAL PASSAGE FOR PIG.
6. DELETED.
7. PIPING CLASS RELEVANT TO INLET TRUNKLINE IS DNV-ST-F101, PIG TRAP IS ACCORDING TO ASME VIII CODE. PIPING CLASS RELEVANT TO PLANT ARE ACCORDING TO ASME B 31.3.
8. PIG DETECTOR NON-INTRUSIVE TYPE.
9. BEND RADIUS REQUIREMENT IS 5D.
10. KICKER PRESSURIZATION AND BALANCE LINE SIZE IS IN LINE WITH STD 03528.ENG.PLI.STD.
11. LOW RANGE PRESSURE INDICATOR TO BE VISIBLE BY OPERATOR WHEN OPERATING VALVES AND DOOR ON PIG TRAPS.
12. DELETED.
13. ZXE 2001 SUPPLIED WITH 011900VR001.
14. REMOVABLE SPOOL FOR POSITIVE ISOLATION SHALL BE FITTED WITH FULLY RATED BLIND FLANGES AT THE LIVE ENDS, IN ACCORDANCE WITH PIPING CLASS.
15. TEMPORARY FACILITIES FOR DEWATERING ACTIVITIES DURING PRECOMMISSIONING TO BE PROVIDED.
16. HIPPS LOGIC SOLVER SUPPLIED BY HIPPS VENDOR.
17. MULTIPHASE FLOW METER.
18. INLET SEALINE DESIGNED FOR 180 MMSCFD @ LOW PRESSURE, 260 MMSCFD @ HIGH PRESSURE.
19. SKIN TEMPERATURE TRANSMITTER TO MONITOR METAL TEMPERATURE DURING PIPELINE DEPRESSURIZATION THROUGH CHOKE VALVE.
20. DELETED
21. MeOH INJECTION IS DISCONTINOUS SERVICE (E.G. DURING START-UP OR AFTER SHUT DOWN) AND IS PERFORMED AS REMEDIAL ACTION ONCE HYDRATE PLUG HAS OCCURRED.
22. DELETED.
23. ACCORDING TO RESPONSABILITY MATRIX DOC 21100BGPQ02103 REV.CDBF09, SECTION 4.1.
24. KEEL ISOLATION VALVE TYPE.
25. FUTURE PROVISION FOR CORROSION INHIBITOR INJECTION.
26. KEEL ISOLATION VALVE TO BE LOCATED AT LEAST AT 3 METERS UNDER WATER IN ALL CONDITIONS.
27. DISTANCE BETWEEN PIG SIGNALLER AND PIG TRAP VALVE SHALL BE MINIMUM OF 150% THE PIG LENGTH.
28. LINE INTERNAL DIAMETER SHALL BE 304.8MM TO ALLOW PIG PASSAGE.EXTERNAL DIAMETER SHALL BE 381.8 MM.
29. TREATED GAS IS USED FOR FLOWLINE FLUSHING AND AS PROPELLING FLUID DURING PIGGING OPERATION.
30. BACKPRESSURE ADJUSTMENT VALVE DURING PIGGING OPERATION PG-0005 TO BE READABLE FROM VALVE.
31. FOR HP PHASE HIPPS SET POINT SHALL BE 120 barg; FOR LP PHASE HIPPS SET POINT TO BE LOWERED TO 75 barg TO PROTECT BOOSTER GAS COMPRESSOR SUCTION.
32. TO BE LOCATED AS CLOSE AS POSSIBLE TO THE RELEVANT INJECTION POINT.
33. PAHH0059 SET POINT (HOLD 140) barg; TO BE LOWERED DURING LP PHASE TO PROTECT BOOSTER GAS COMPRESSORS SUCTION.
34. HIPPS CLOSURE TIME (HOLD 140) sec.
35. SDV-0201 CLOSURE TIME IS (HOLD 140) sec AND SDV-0202 CLOSURE TIME IS (HOLD 140) sec.
36. 011900PAHH0002 SETPOINT IS FIXED AT 250 barg AND IT IS USED TO INHIBIT THE OPENING OF 011900SDV0201 ABOVE THE SETPOINT. 011900PAH0002 SETPOINT IS FIXED AT 110 barg AND IT IS USED TO INHIBIT THE OPENING OF 011900SDV0202 ABOVE THE SETPOINT.
37. 011900TI0001, 011900FI0036 AND 011900PI0059 TO BE READABLE FROM CHOKE VALVE 011900RA454 AND ANGLE VALVE 011900RA132.
38. QUICK COUPLING CONNECTION FOR LIQUID DISPOSAL.
39. 015400RB0162 IS NORMALLY OPENED. IT IS TO BE CLOSED DURING PIGGING TO AVOID SENDING THE SLURRY TO HAZARDOUS OPEN DRAIN.
40. 015400RB0161 IS NORMALLY CLOSED. IT IS TO BE OPENED DURING PIGGING.
41. MAJOR BARREL TO BE PROVIDED WITH INTERNAL BASKET.
42. HIPPS PTs SENSORS ARE INTERLOCKED BY RELEVANT SINGLE MANIFOLD (no.1 FOR EACH PT). OPENING/CLOSURE OF RELEVANT MANIFOLD SHALL BE BASED ON PRE-DETERMINATED SAFE SEQUENCE REPORTED IN RELEVANT WORKING/OPERATING INSTRUCTION. HIPPS 2oo3 VOTING SHALL REVERT TO A 1oo2 VOTING LOGIC DURING SINGLE SENSOR MAINTENANCE OR FAULT DETECTED.
43. SPECIAL COATING USED FOR LINE 011900WF7962-16"-98P-V/B005 SHALL BE APPLIED FROM ESDV-0101 TO RISER BALCONY FLOOR LEVEL, WHERE THE LINE IS ACCESSIBLE FOR MAINTENANCE.
44. CHECK VALVE SHALL BE DISSIMILAR TYPE FROM THE CHECK VALVE 011200RC0420 IN THE CONTINUATION P&ID 220901DPFM03862.
45. SWIVEL FLANGE WITH EXTERNAL TEST PORT.

HOLDS:

ID	HOLD SUBJECT	DEPT IN CHARGE OF HOLD RESOLUTION	DEPT IN CHARGE OF P&ID UPDATE
60	NUMBER AND LOCATION OF SKIN TEMPERATURE INSTRUMENT IS ON HOLD.	BY PROCESS DEPT.	BY PROCESS DEPT.
69	REQUIREMENT FOR CHOKE VALVE BYPASS TO BE EVALUATED BASED ON FLOW ASSURANCE INFORMATION.	BY PROCESS DEPT.	BY PROCESS DEPT.
87	SOLVED		
103	DETAILS RELEVANT TO THE HYDRAULIC POWER UNIT 014500XX002 ARE ON HOLD PENDING ON ENGINEERING DEVELOPMENT.	BY INSTRUMENT DEPT.	BY PROCESS DEPT.
	SOLVED		
139	LINE SIZE FOR PIGGING OPERATION WILL BE DEFINED PENDING ON FLOW ASSUNRANCE DATA.	BY PROCESS DEPT.	BY PROCESS DEPT.
140	HIPPS CLOSURE TIME, SDV CLOSURE TIME AND PAHH SET POINT TO BE DEFINED PENDING ON DYNAMIC ANALYSIS	BY PROCESS DEPT.	BY PROCESS DEPT.
582	PACKAGE CONTROL SYSTEM WITH RELEVANT NUMBER OF SIGNALS IS ON HOLD, PENDING ON VENDOR INFORMATION.	BY INSTRUMENT DEPT.	BY INSTRUMENT DEPT.

011900VR001		GEHEM CLUSTER NORTH-LAUNCHING/ RECEIVING TRAP
SERVICE		
NOMINAL DIAMETER (MINOR/MAJOR)	in	16" / 18"
TOTAL LENGTH/HEIGHT	mm	11200
DESIGN PRESSURE	barg	543
DESIGN TEMPERATURE	°C	75 / -35
INSULATION	Yes/No	No
TRIM LINE		011900TL0785-97P-I/B004

Validity Status	Revision Number	Date	Description	Prepared by	Checked by	Approved by	Contractor Approved	Company Approved
EX-DE	01	19/12/2025	ISSUED FOR CONSTRUCTION	G. Cecchelli	V.Palazzolo	L.Manzi	N/A	N/A
EX-DE	00	25/04/2025	ISSUED FOR CONSTRUCTION	I. Onwubuariri	V.Palazzolo	L.Manzi	N/A	N/A
CD-FE	00	05/11/2024	ISSUED FOR DESIGN	PAMO	VAPA	STBI	N/A	N/A

Revision Index

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Facility and Sub Facility description: Kutei North Hub - FPSO TOPSIDE

Project and SoW Description: Indonesia Kutei North Hub Project FPSO Package

Scale: N/A

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Document Title: PIPING & INSTRUMENTATION DIAGRAM
UNIT 190 - GEHEM CLS NORTH LAUNCHING - RECEIVING TRAP

Supersedes N.

Superseded by N.

Plant Area: N/A

Functional Unit: 190

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